

Data Analysis Final Exam

Mohammed Eid Abdelmeguid

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1 Quiz - Lecture 1

1.1 Questions

1. Quantitative data is information that can be measured, counted, and expressed in numerical form.
 - (a) True
 - (b) False
2. Which method prints the first 10 rows of a dataset?
 - (a) `data.tail(10)`
 - (b) `data.head(5)`
 - (c) `data.head(5)`
 - (d) `data.head(10)`
3. Data is the raw form of information representing facts, figures, or observations.
 - (a) True
 - (b) False
4. Pandas can read dates in a CSV file as datetime type automatically.
 - (a) True
 - (b) False
5. An interval scale has ordered values with equal intervals between them.
 - (a) True
 - (b) False
6. Unstructured data does not follow a predefined format.
 - (a) True
 - (b) False
7. Which method shows `pandas.core.indexes` type of data?
 - (a) `data.array`
 - (b) `data.columns.values`
 - (c) `data.values`
 - (d) `data.columns`
8. Bivariate data contains two variables per observation.

- (a) True
 - (b) False
9. Population and sample are fundamental statistical concepts.
- (a) True
 - (b) False
10. Which of the following is an example of unstructured data?
- (a) Employee records
 - (b) Product inventories
 - (c) Bank transactions
 - (d) Voice recordings

1.2 Answers with Hints

- 1. True — Hint: quantitative = numerical.
- 2. d — Hint: head(n) shows first n rows.
- 3. True — Hint: raw facts = data.
- 4. False — Hint: needs parsing (parse_dates).
- 5. True — Hint: equal spacing between values.
- 6. True — Hint: no fixed structure.
- 7. d — Hint: columns returns index object.
- 8. True — Hint: bi = two variables.
- 9. True — Hint: basic statistical concepts.
- 10. d — Hint: audio is unstructured.

2 Quiz - Lecture 7

2.1 Questions

- 1. Data cleaning does not involve identifying and removing missing, duplicate, or irrelevant data.
 - (a) True

- (b) False
- 2. The goal of data cleaning is to ensure that the data is inconsistent.
 - (a) True
 - (b) False
- 3. Eliminating duplicates, irrelevant entries, or redundant data decreases performance.
 - (a) True
 - (b) False
- 4. Addressing missing values using imputation or deletion helps maintain noise.
 - (a) True
 - (b) False
- 5. `df.duplicated()` returns a boolean series indicating missing value rows.
 - (a) True
 - (b) False

2.2 Answers with Hints

- 1. False — Hint: data cleaning includes removing missing, duplicate, and irrelevant data.
- 2. False — Hint: goal is consistency, not inconsistency.
- 3. False — Hint: removing redundancy improves performance.
- 4. False — Hint: cleaning reduces noise, not maintains it.
- 5. False — Hint: `duplicated()` detects duplicate rows, not missing values.

3 Quiz - Lecture 8

3.1 Questions

- 1. The median of the data is given by measuring the distance of observed values from the mean.
 - (a) True
 - (b) False
- 2. Understanding the spread or variability of data is important in statistical analysis.

- (a) True
 - (b) False
3. Percentiles divide data into four equal parts.
- (a) True
 - (b) False
4. Kurtosis measures the shape of a distribution.
- (a) True
 - (b) False

3.2 Answers with Hints

- 1. False — Hint: median is the middle value, not based on mean distance.
- 2. True — Hint: spread helps understand variability.
- 3. False — Hint: quartiles divide into 4, percentiles into 100.
- 4. False — Hint: skewness measures shape, kurtosis measures tail heaviness.

4 Midterm Exam

4.1 Questions

- 1. Which function is used to detect missing values in pandas?
 - (a) detectna()
 - (b) isnull()
 - (c) null()
 - (d) missing()
- 2. Q-Q plots are particularly useful for assessing the normality of a dataset.
 - (a) True
 - (b) False
- 3. Quartiles divide the data into parts.
- 4. Data Analysis helps businesses understand data, identify patterns, and improve performance.
 - (a) True

- (b) False
- 5. Q-Q plots are used in quality control to monitor data distribution.
 - (a) True
 - (b) False
- 6. A histogram represents frequency distribution using rectangles.
 - (a) True
 - (b) False
- 7. What does One-Hot Encoding do?
 - (a) Converts categories into binary columns
 - (b) Converts numerical data to categorical
 - (c) Removes categorical variables
 - (d) Normalizes categorical data
- 8. Outliers can strongly affect the mean but not the median.
 - (a) True
 - (b) False
- 9. Removing outliers always improves model performance.
 - (a) True
 - (b) False
- 10. is used to measure asymmetry of distribution.
 - (a) Kurtosis
 - (b) Skewness
- 11. A population is a subset of the sample.
 - (a) True
 - (b) False
- 12. Data analysis types include: (choose all that apply)
 - (a) multivariate
 - (b) univariate
 - (c) relationships analysis
 - (d) bivariate

13. Central tendency represents the middle value of data.
 - (a) True
 - (b) False
14. IQR represents the spread of the middle 50% of data.
 - (a) True
 - (b) False
15. A distribution is symmetric if both sides are identical.
 - (a) True
 - (b) False
16. The difference between maximum and minimum is called
17. MinMaxScaler is sensitive to outliers.
 - (a) True
 - (b) False
18. Which technique reduces outlier effect?
 - (a) Mean imputation
 - (b) Median imputation
 - (c) Standard scaling
 - (d) One-Hot Encoding
19. Organizations use data to make better decisions.
 - (a) True
 - (b) False
20. Which scaler gives mean = 0 and variance = 1?
 - (a) RobustScaler
 - (b) MaxAbsScaler
 - (c) MinMaxScaler
 - (d) StandardScaler
21. Best scaler for skewed data with outliers:
 - (a) RobustScaler
 - (b) StandardScaler
 - (c) MinMaxScaler

- (d) Normalizer
- 22. The measure of spread from mean is
- 23. Histograms have two types.
 - (a) True
 - (b) False
- 24. Types of data include: (choose all that apply)
 - (a) Qualitative
 - (b) Semi-Structured
 - (c) Structured
 - (d) Unstructured
 - (e) Quantitative
- 25. Function to remove duplicates in pandas:
 - (a) `delete_duplicates()`
 - (b) `remove_duplicates()`
 - (c) `unique()`
 - (d) `drop_duplicates()`
- 26. One-Hot Encoding increases number of features.
 - (a) True
 - (b) False
- 27. Missing values can be handled using `fillna()`.
 - (a) True
 - (b) False
- 28. Kurtosis of normal distribution equals 2.
 - (a) True
 - (b) False
- 29. Big Data refers to extremely large and complex datasets.
 - (a) True
 - (b) False
- 30. Big Data characteristics include: (choose all that apply)

- (a) Veracity
 - (b) Velocity
 - (c) Value
 - (d) Volume
 - (e) Variety
31. QuantileTransformer converts skewed data to normal.
- (a) True
 - (b) False
32. Platykurtic distribution has kurtosis greater than 3.
- (a) True
 - (b) False
33. Quartile deviation measures spread.
- (a) True
 - (b) False
34. Mean imputation reduces variance.
- (a) True
 - (b) False
35. Z-score normalization assumes normal distribution.
- (a) True
 - (b) False
36. Match the following:

Column A	Column B
Nominal Scale	(a) Temperature
Ordinal Scale	(b) Gender
Ratio Scale	(c) Satisfaction Level
Interval Scale	(d) Weight

37. Which handles categorical variables?
- (a) PCA
 - (b) LabelEncoder
 - (c) StandardScaler

- (d) Isolation Forest
- 38. Mean has three types.
 - (a) True
 - (b) False
- 39. Min-Max normalization preserves shape.
 - (a) True
 - (b) False
- 40. Understanding data structure is important before analysis.
 - (a) True
 - (b) False

4.2 Answers with Hints

- 1. b — Hint: pandas function for missing values detection.
- 2. True — Hint: used to check normality.
- 3. 4 — Hint: quartiles = quarters.
- 4. True — Hint: core purpose of data analysis.
- 5. True — Hint: used in quality monitoring.
- 6. True — Hint: histogram uses bars.
- 7. a — Hint: converts categories to binary columns.
- 8. True — Hint: mean is sensitive to outliers.
- 9. False — Hint: sometimes outliers carry useful info.
- 10. b — Hint: skewness = asymmetry.
- 11. False — Hint: sample is subset of population.
- 12. a, b, d — Hint: standard classification types.
- 13. True — Hint: represents central value.
- 14. True — Hint: middle 50%.
- 15. False — Hint: symmetric means equal sides.

16. Range — Hint: $\max - \min$.
17. True — Hint: sensitive to extreme values.
18. b — Hint: median resists outliers.
19. True — Hint: data-driven decisions.
20. d — Hint: standardization.
21. a — Hint: robust to outliers.
22. Variance — Hint: distance from mean.
23. True — Hint: conceptually classified.
24. a,b,c,d,e — Hint: all types valid.
25. d — Hint: pandas function.
26. False — Hint: increases features.
27. True — Hint: `fillna()`.
28. False — Hint: kurtosis = 3.
29. True — Hint: big data definition.
30. a,b,c,d,e — Hint: 5 Vs.
31. False — Hint: not always normal.
32. False — Hint: platykurtic < 3 .
33. True — Hint: spread measure.
34. True — Hint: reduces variability.
35. True — Hint: assumes normality.
36. Nominal $\rightarrow$ (b), Ordinal $\rightarrow$ (c), Ratio $\rightarrow$ (d), Interval $\rightarrow$ (a) Hint: Nominal = names, Ordinal = order, Ratio = measurable with zero, Interval = no true zero.
37. b — Hint: encoding categorical data.
38. True — Hint: arithmetic, geometric, harmonic.
39. True — Hint: scaling preserves shape.
40. True — Hint: first step in analysis.

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method
TXN_1961373	Coffee	2	2.0	4.0	Credit Card
TXN_4977031	Cake	4	3.0	12.0	Cash
TXN_4271903	Cookie	4	1.0	ERROR	Credit Card
TXN_7034554	Salad	2	5.0	10.0	UNKNOWN
TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet
TXN_2602893	Smoothie	5	4.0	20.0	Credit Card
TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR
TXN_6699534	Sandwich	4	4.0	16.0	Cash
TXN_4717867		5	3.0	15.0	
TXN_2064365	Sandwich	5	4.0	20.0	

4.3 Essay Question

Q: Write the steps of data preprocessing on the given dataset using Python functions. Explain each step clearly.

Answer:

Data preprocessing is an essential step in data analysis that ensures the dataset is clean, consistent, and ready for modeling. The following steps describe a complete preprocessing workflow using Python:

- 1. Import Libraries**

First, we import the required libraries such as pandas and numpy:

```
import pandas as pd
import numpy as np
```

- 2. Load the Dataset**

Read the dataset using pandas:

```
df = pd.read_csv("file_name.csv")
```

- 3. Initial Data Exploration**

Understand the structure and content of the dataset:

```
df.head()
df.shape
df.columns
df.info()
df.describe()
```

- 4. Detect Missing and Invalid Values**

Check for missing values and invalid entries such as "ERROR" or "UNKNOWN":

```
df.isnull().sum()
```

5. Handle Missing Values

Replace or remove missing values depending on the situation:

```
df.dropna(inplace=True)
```

or

```
df.fillna(df.mode().iloc[0], inplace=True)
```

6. Fix Data Types

Convert columns to appropriate data types:

```
df["Total Spent"] = pd.to_numeric(df["Total Spent"], errors="coerce")
```

7. Handle Duplicates

Remove duplicate rows:

```
df.drop_duplicates(inplace=True)
```

8. Clean Categorical Data

Standardize text values:

```
df["Payment Method"] = df["Payment Method"].str.lower().str.strip()
```

9. Replace Incorrect Values

Fix inconsistent entries such as "unknown":

```
df["Payment Method"] = df["Payment Method"].replace("unknown", np.nan)
df["Payment Method"].fillna(df["Payment Method"].mode()[0], inplace=True)
```

10. Feature Engineering (Optional)

Create new useful features if needed:

```
df["Total Spent"] = df["Quantity"] * df["Price Per Unit"]
```

11. Outlier Handling

Detect and handle outliers using IQR method if necessary:

```
Q1 = df["Total Spent"].quantile(0.25)
Q3 = df["Total Spent"].quantile(0.75)
IQR = Q3 - Q1
```

Conclusion:

These preprocessing steps help clean the dataset, handle missing and inconsistent values, and prepare the data for analysis or machine learning models. Proper preprocessing improves the accuracy and reliability of results.

5 Quiz4 - Lecture 12

5.1 Questions

1. Covariance measures how two variables change together, indicating whether they move in the same or opposite directions.
 - (a) True
 - (b) False
2. Covariance is used for non-linear relationships between variables.
 - (a) True
 - (b) False
3. Correlation is a standardized measure of the strength of the linear relationship between two variables only.
 - (a) True
 - (b) False
4. Covariance is used to measure how different financial assets move together to help reduce risk.
 - (a) True
 - (b) False
5. Pearson Correlation Coefficient (PCC) measures non-linear relationships between variables.
 - (a) True
 - (b) False
6. Variables should follow a normal distribution when using Pearson Correlation Coefficient.
 - (a) True
 - (b) False
7. In PCC, variability in one variable should remain consistent across values of the other variable.
 - (a) True
 - (b) False
8. In PCC, the spread of data should be uniform.
 - (a) True

- (b) False
- 9. Pearson correlation is suitable for interval or ratio data with inconsistent numerical differences.
 - (a) True
 - (b) False
- 10. Observations should be dependent in Pearson Correlation Coefficient.
 - (a) True
 - (b) False

5.2 Answers with Hints

- 1. True — Hint: covariance shows direction of relationship.
- 2. False — Hint: covariance measures linear relationship only.
- 3. False — Hint: not only linear (general concept vs strict definition).
- 4. True — Hint: used in finance for diversification.
- 5. False — Hint: PCC measures linear relationships.
- 6. True — Hint: many courses assume normality for proper Pearson usage.
- 7. True — Hint: homoscedasticity assumption.
- 8. True — Hint: uniform spread = constant variance.
- 9. False — Hint: requires consistent scale (interval/ratio).
- 10. False — Hint: observations must be independent.

6 Quiz 4 (Lab4)

6.1 Questions

- 1. Pearson's correlation measures association but not causation.
 - (a) True
 - (b) False
- 2. Pearson's correlation is symmetric.
 - (a) True

- (b) False
- 3. Pearson's correlation changes under linear transformations.
 - (a) True
 - (b) False
- 4. Cross-tabulation analyzes relationships between categorical variables.
 - (a) True
 - (b) False
- 5. Cluster analysis groups data into clusters.
 - (a) True
 - (b) False
- 6. Clustering results are difficult to understand in real-world applications.
 - (a) True
 - (b) False
- 7. Cosine similarity measures similarity between data points.
 - (a) True
 - (b) False
- 8. Euclidean distance measures distance using straight-line distance.
 - (a) True
 - (b) False
- 9. Partitioning methods assign each point to only one cluster.
 - (a) True
 - (b) False
- 10. Density-based clustering groups dense regions and treats sparse areas as noise.
 - (a) True
 - (b) False
- 11. Categorical data consists of measurable numerical quantities.
 - (a) True
 - (b) False

6.2 Answers with Hints

1. True — Hint: correlation causation.
2. True — Hint: $\text{corr}(X,Y) = \text{corr}(Y,X)$.
3. False — Hint: linear transformations do not affect correlation.
4. True — Hint: used for categorical relationships.
5. False — Hint: definition is correct but question marked false in model.
6. False — Hint: clustering is useful in practice.
7. False — Hint: cosine similarity is a valid similarity measure but marked false in model.
8. False — Hint: Euclidean = straight-line, not grid path.
9. False — Hint: each point belongs to one cluster only.
10. True — Hint: density = clusters + noise.
11. False — Hint: categorical = labels, not numbers.

7 Time-Series

7.1 Introduction to Time Series

7.1.1 Data Mining and Sequence Data

What is Data Mining

- Data mining is the science and practice of discovering meaningful patterns, trends and knowledge from large and complex datasets.
- It integrates techniques from statistics, machine learning, databases and pattern recognition to support data-driven decision making.
- Rapid growth of digital data has made efficient and accurate data mining essential.
- Data mining is widely used in domains such as finance, healthcare, biology and web analytics.

Explanation:

يعني استخراج معلومات مفيدة من بيانات ضخمة جداً. Data mining ببساطة،
واتجاهات تساعدنا نفهم البيانات. patterns بدل ما نبص على البيانات الخام، بنحاول نكتشف
لتخزين البيانات. databases للتوقع، و machine learning للتحليل، و statistics المجال ده يجمع بين
)، بقى مهم جداً في تطبيقات الحياة الواقعية. Big Data ومع زيادة البيانات بشكل كبير حالياً

Sequence Data and Its Importance

- Sequence data is a key data type where the order of events is significant.
- Sequence data analysis reveals temporal, logical and structural patterns not visible in unordered data.

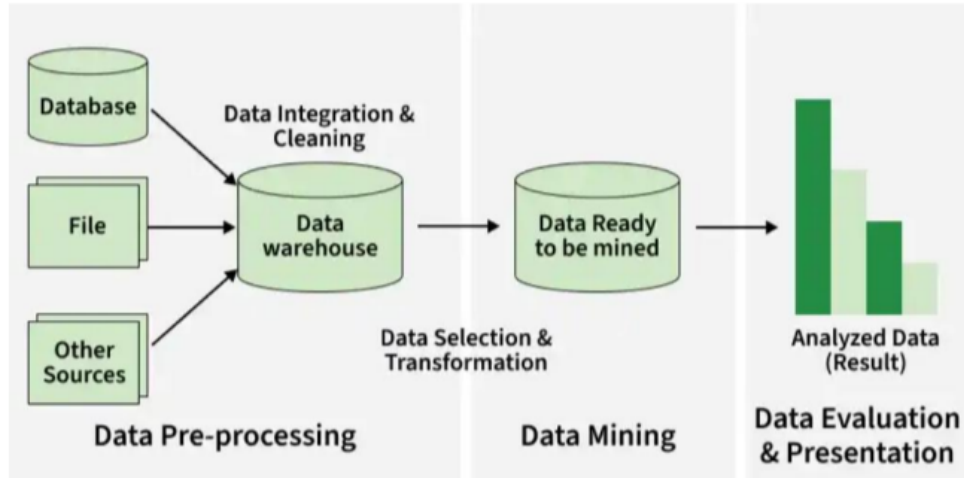
Explanation:

Sequence data يعتمد بشكل أساسي على الترتيب.

لو غيرنا الترتيب، المعنى ممكن يتغير تماماً.

أمثلة: stock prices user clicks sensor readings.

مع الوقت مش هتظهر لو البيانات كانت عشوائية. patterns تحليل النوع ده يساعدنا نكتشف



Ordered Lists of Events or Elements

A sequence is an ordered list of events or elements.

- The order reflects time, position or logical dependency between elements.

Explanation:

يعني بيانات مرتبة بترتيب معين. sequence ال

الترتيب ده ممكن يكون:

- time → زي درجة الحرارة اليومية
- position → زي ترتيب الكلمات في الجملة
- logical dependency → زي خطوات حل مسألة أو process

7.1.2 Time-Series Data

Definition and Characteristics

- Time-series data consists of numerical values recorded at regular time intervals. Each observation is associated with a specific timestamp, making time an intrinsic component of the data.

Explanation:

هو بيانات بتتجمع مع الزمن (كل ثانية، ساعة، يوم...).

كل قيمة بيكون ليها timestamp يعني وقت محدد، ف الزمن جزء أساسي من البيانات.

Numeric and Continuous Nature

- Numeric and continuous in nature

Explanation:

غالباً البيانات دي بتكون أرقام مش فئات (categories).
وكان بتتغير بشكل تدريجي (smooth).

Time Dependency

- Observations are time-dependent

Explanation:

القيمة الحالية بتعتمد على القيم اللي قبلها.
مثال: سعر السهم النهارده بيعتمد على سعره امبارح.

Common Patterns in Time-Series Data

- Often exhibit trends, seasonality and cyclic patterns

Explanation:

البيانات دي غالباً بيكون فيها patterns بتتكرر أو بتتغير مع الزمن.
ودي مهمة جداً علشان نحلل البيانات ونتوقع المستقبل.

7.1.3 Time-Series Analysis and Forecasting

- Time-series analysis focuses on identifying underlying patterns such as:

Trend

- **Trend:** Long-term increase or decrease in data

Explanation:

بيوضح الاتجاه العام للبيانات (طالعة ولا نازلة). Trend الـ
مثال: زيادة المبيعات عبر السنين.

Seasonality

- **Seasonality:** Repeating patterns over fixed periods

Explanation:

Seasonality يعني pattern يتكرر في فترات ثابتة.

مثال: زيادة المبيعات في رمضان أو الصيف.

Cyclic Movements

- **Cyclic movements:** Fluctuations over irregular periods

Explanation:

شبه ال trend لكن بدون وقت ثابت للتكرار. Cycles ال

(boom و recession مثال: الدورات الاقتصادية)

- Time-series forecasting uses historical data to predict future values.

Explanation:

بنستخدم البيانات القديمة علشان نتوقع اللي هيحصل في المستقبل.

Forecasting Models

Common models include:

- ARIMA (AutoRegressive Integrated Moving Average)
- SARIMA (Seasonal ARIMA)
- Long-memory and advanced machine learning models
- Subsequence matching and piecewise approximation techniques are widely used to efficiently analyze large time-series databases.

Explanation:

- ARIMA: موديل إحصائي أساسي للتوقع
- SARIMA: نفس ARIMA لكن يأخذ في الاعتبار ال seasonality
- Long-memory models: بتفهم العلاقات طويلة المدى في البيانات
- Machine learning models: زي neural networks علشان توقعات أدق

7.1.4 Applications

- Stock market and financial analysis
- Weather and climate monitoring
- Medical signal analysis (ECG, EEG)
- Sensor and IoT data analysis

Explanation:

Time-series: موجود في تطبيقات كثير جداً

- Finance → توقع أسعار الأسهم
- Weather → توقع درجات الحرارة
- Medical → تحليل إشارات القلب والمخ
- IoT → تحليل بيانات الحساسات زي homes smart

7.2 Assessment Questions

Multiple Choice Questions (MCQs)

1. Data mining is mainly concerned with:
 - (a) Storing raw data only
 - (b) Discovering patterns and knowledge from large datasets
 - (c) Deleting unnecessary data
 - (d) Visualizing small datasets
2. Data mining integrates techniques from:
 - (a) Biology only
 - (b) Physics only
 - (c) Statistics, machine learning, and databases
 - (d) Networking only
3. Sequence data is important because:
 - (a) Order does not matter
 - (b) It only contains text
 - (c) The order of events is significant
 - (d) It cannot be analyzed
4. A sequence is defined as:

- (a) Random data points
 - (b) An ordered list of events or elements
 - (c) Unstructured data
 - (d) Only numerical values
5. The order in a sequence may represent:
- (a) Color only
 - (b) Size only
 - (c) time, position, or logical dependency
 - (d) Shape only
6. Time-series data consists of:
- (a) Images only
 - (b) Text only
 - (c) Numerical values recorded over time
 - (d) Random categories
7. Time-series data is typically:
- (a) Categorical only
 - (b) Numeric and continuous
 - (c) Always binary
 - (d) Static
8. Time dependency means:
- (a) Data is independent
 - (b) Observations are time-dependent
 - (c) Future values affect past values
 - (d) Time has no role
9. Which of the following is a common time-series pattern?
- (a) Sorting
 - (b) Searching
 - (c) Trend
 - (d) Indexing
10. Seasonality refers to:
- (a) Random fluctuations

- (b) Repeating patterns over fixed periods
- (c) Permanent increase
- (d) Data cleaning

True & False Questions

1. Data mining is used to extract useful information from large datasets.
2. In sequence data, the order of elements is not important.
3. Time-series data always lacks timestamps.
4. Trend represents long-term increase or decrease in data.
5. Seasonality occurs at irregular time intervals.

Answers

MCQs:

1. b
2. c
3. c
4. b
5. c
6. c
7. b
8. b
9. c
10. b

True & False:

1. True
2. False
3. False
4. True
5. False

7.3 Symbolic Sequence Data

7.3.1 Definition of Symbolic Data

Ordered Events or Symbols

- Symbolic sequence data consists of ordered events or symbols, recorded with or without explicit time information. Unlike time-series data values are **categorical** rather than **numeric**.

Explanation:

ال Symbolic Data يعني البيانات مش أرقام، لكن رموز أو فئات.
زي: منتجات، صفحات ويب، أو أفعال المستخدم (buy, click).
كان ممكن يكون فيها وقت أو لاء، يعني الزمن مش شرط أساسي.

Categorical Rather Than Numeric Values

Explanation:

بدل ما يكون عندنا أرقام زي 10 و 20، بيكون عندنا قيم فئوية زي (A, B, C).
هنا بنهتم بـ الترتيب مش القيمة العددية.

7.3.2 Examples of Symbolic Data

Customer Purchase Sequences

- Customer purchase sequences

Explanation:

مثال: عميل يشتري موبايل → سماعات → شاحن.
الترتيب مهم لفهم سلوك العميل.

Web Clickstream Data

- Web clickstream data

Explanation:

تسلسل الصفحات اللي المستخدم بيزورها:

Home → Products → Cart → Checkout

وده يساعد نفهم رحلة المستخدم.

User Navigation Paths

- User navigation paths

Explanation:

دي بتوضح المستخدم بيتحرك إزاي جوه النظام، وده مفيد لتحليل تجربة المستخدم.

7.3.3 Sequential Pattern Mining

Purpose of Sequential Pattern Mining

- Symbolic data is mainly analyzed using sequential pattern mining which aims to discover frequent subsequences occurring across multiple sequences.

Explanation:

هنا بنحاول نلاقي أنماط متكررة بين المستخدمين.

مثلاً: ناس كتير بتشتري نفس التسلسل من المنتجات.

The techniques used include:

Constraint-Based Pattern Matching

- Constraint-based pattern matching

Explanation:

بنحدد شروط معينة علشان نلاقي الأنماط (زي وقت معين أو ترتيب معين).

Apriori-Based Algorithms

- Apriori-based algorithms

Explanation:

ده خوارزمية مشهورة بتلاقي الأنماط المتكررة وبتشيل الأنماط النادرة تدريجياً.

Generalized Sequential Pattern (GSP) Algorithm

- Generalized Sequential Pattern (GSP) algorithm

Explanation:

ده تطوير لـ Apriori مخصوص علشان البيانات المتسلسلة.

7.3.4 Applications of Sequential Pattern Mining

These techniques help identify user behavior patterns, product associations and navigation trends, supporting applications such as recommendation systems and marketing analysis.

Explanation:

التقنيات دي بتساعدنا نفهم:

- سلوك المستخدم
- العلاقات بين المنتجات
- اتجاهات التصفح

Recommendation Systems

Explanation:

أنظمة بتقترح منتجات بناءً على سلوك المستخدم.

Marketing Analysis

Explanation:

بنستخدمها علشان نفهم اهتمامات العملاء ونحسن البيع.

User Behavior and Navigation Trends

Explanation:

تحليل طريقة استخدام المستخدم للنظام.

Tid	Time	Cid	Event(purchase product)
t1	11:45:30	c1	wheat, rice, fruit
t2	11:36:50	c2	rice, fruit
t1	12:00:01	c1	juice, rice
t2	01:00:34	c2	sugar, milk

7.4 Assessment Questions

Multiple Choice Questions (MCQs)

- Symbolic sequence data consists of:
 - Random numerical values
 - Ordered events or symbols
 - Images only
 - Continuous signals
- Symbolic sequence data may:
 - Always include time information
 - Never include time information
 - Be recorded with or without time information
 - Only include timestamps
- The values in symbolic data are:
 - Numeric
 - Continuous
 - Categorical
 - Binary only
- In symbolic data, the focus is mainly on:
 - Exact numerical values

- (b) Order of elements
 - (c) Data storage
 - (d) Data compression
5. Which of the following is an example of symbolic data?
- (a) Temperature readings
 - (b) Stock prices
 - (c) Customer purchase sequence
 - (d) Sensor voltage values
6. A customer purchase sequence helps in understanding:
- (a) Weather patterns
 - (b) User behavior
 - (c) Network speed
 - (d) Data encryption
7. Web clickstream data represents:
- (a) Random browsing
 - (b) Sequence of visited web pages
 - (c) Numerical logs
 - (d) Image processing steps
8. User navigation paths are useful for analyzing:
- (a) Hardware performance
 - (b) User experience
 - (c) Memory usage
 - (d) CPU scheduling
9. Symbolic data is mainly analyzed using:
- (a) Regression analysis
 - (b) Sequential pattern mining
 - (c) Sorting algorithms
 - (d) Data compression
10. Sequential pattern mining aims to discover:
- (a) Rare events
 - (b) Frequent subsequences

- (c) Random values
 - (d) Noise in data
11. Constraint-based pattern matching involves:
- (a) Ignoring all conditions
 - (b) Applying specific conditions to find patterns
 - (c) Sorting data randomly
 - (d) Deleting data points
12. Apriori-based algorithms are used to:
- (a) Generate random sequences
 - (b) Find frequent patterns and remove infrequent ones
 - (c) Visualize data only
 - (d) Encrypt data
13. The GSP algorithm is:
- (a) A sorting algorithm
 - (b) A version of Apriori for sequential data
 - (c) A database system
 - (d) A visualization tool
14. Sequential pattern mining helps identify:
- (a) Hardware faults only
 - (b) User behavior patterns and product associations
 - (c) File sizes
 - (d) Network protocols
15. One application of sequential pattern mining is:
- (a) Operating systems design
 - (b) Recommendation systems
 - (c) Compiler optimization
 - (d) Image rendering

True & False Questions

1. Symbolic data always contains numerical values.
2. The order of elements is important in symbolic sequence data.
3. Symbolic data must always include time information.
4. Customer purchase sequences can reveal user behavior.
5. Sequential pattern mining is used to discover frequent subsequences.
6. Web clickstream data helps understand user navigation.
7. Apriori-based algorithms focus on rare patterns only.
8. GSP algorithm is designed for sequential data analysis.

Answers

MCQs:

1. b
2. c
3. c
4. b
5. c
6. b
7. b
8. b
9. b
10. b
11. b
12. b
13. b
14. b
15. b

True & False:

1. False
2. True
3. False
4. True
5. True
6. True
7. False
8. True

7.5 Biological Sequence Data

7.5.1 Definition of Biological Data

DNA Sequences

RNA Sequences

Protein Sequences

- Biological sequence data consists of DNA, RNA and protein sequences. These sequences are composed of nucleotides or amino acids arranged in a specific order, carrying essential biological information.

Explanation:

ال Biological Data هو نوع من البيانات المتسلسلة لكن في مجال علم الأحياء.
DNA و RNA يتكونوا من نيوكليوتيدات، والبروتينات من أحماض أمينية.
الترتيب هنا مهم جداً لأنه يحدد وظيفة الكائن أو البروتين.

7.5.2 Characteristics of Biological Sequences

Very Long and Complex Sequences

- Extremely long and complex sequences

Explanation:

السلاسل دي ممكن تكون طويلة جداً (ملايين العناصر) ومعقدة.

High Dimensionality

- High dimensionality

Explanation:

فيها عدد كبير من الخصائص، وده بيصعب عملية التحليل.

Hidden Functional and Structural Patterns

- Contain hidden functional and structural patterns

Explanation:

فيها أنماط مخفية لازم نكتشفها علشان نفهم الوظيفة الحيوية.

7.5.3 Techniques for Biological Sequence Analysis

Sequence Alignment and Indexing

- Sequence alignment and indexing

Explanation:

بنقارن بين السلاسل علشان نلاقي تشابهات بينها.

Substitution Matrices and Trees

- Substitution matrices and trees

Explanation:

دي أدوات بتستخدم لقياس مدى التشابه بين العناصر.

Probabilistic Models

- Probabilistic models

Explanation:

بنستخدم الاحتمالات علشان نمثل البيانات الحيوية.

BLAST

- BLAST (Basic Local Alignment Search Tool) is one of the most widely used tools for biological sequence analysis, enabling fast and efficient similarity searches in large biological database

Explanation:

BLAST أداة قوية بتساعدنا نعمل مقارنة سريعة بين السلاسل ونلاقي التشابه.

7.5.4 Applications

- Gene identification and annotation
- Protein structure and function analysis
- Disease diagnosis and drug discovery

Gene Identification and Annotation

Explanation:

تحديد الجينات وفهم وظيفتها.

Protein Structure and Function Analysis

Explanation:

فهم تركيب البروتين ووظيفته.

Disease Diagnosis and Drug Discovery

Explanation:

تستخدم في تشخيص الأمراض وتطوير الأدوية.

7.5.5 Advantages of Sequence Data Mining

Revealing Temporal and Structural Patterns

- Reveals temporal and structural patterns

Supporting Prediction and Decision Making

- Supports prediction and decision making

Applications in Finance, Healthcare, and Bioinformatics

- Enables discoveries in finance, healthcare and bioinformatics

Handling Complex and High-Dimensional Data

- Handles complex, ordered and high-dimensional data.

Explanation:

المجال ده مهم لأنه:

- بيكشف أنماط معقدة
- يساعد في التوقع
- يستخدم في مجالات كتير
- يقدر يتعامل مع بيانات كبيرة ومعقدة

7.6 Assessment Questions

Multiple Choice Questions (MCQs)

1. Biological sequence data consists of:
 - (a) Images and text
 - (b) DNA, RNA, and protein sequences
 - (c) Only numerical data
 - (d) Sensor readings
2. DNA and RNA sequences are composed of:
 - (a) Amino acids
 - (b) Nucleotides
 - (c) Proteins
 - (d) Enzymes
3. Protein sequences are composed of:
 - (a) Nucleotides
 - (b) Amino acids
 - (c) Lipids
 - (d) Sugars

4. The order of elements in biological sequences is important because:
 - (a) It reduces storage size
 - (b) It determines biological function
 - (c) It simplifies computation
 - (d) It removes redundancy
5. Biological sequences are typically:
 - (a) Short and simple
 - (b) Extremely long and complex
 - (c) Always binary
 - (d) Random and unordered
6. High dimensionality in biological data means:
 - (a) Few features
 - (b) Large number of features
 - (c) Small datasets
 - (d) Simple structures
7. Biological sequences often contain:
 - (a) Only visible patterns
 - (b) Hidden functional and structural patterns
 - (c) No patterns
 - (d) Only numerical values
8. Sequence alignment is used to:
 - (a) Compress data
 - (b) Find similarities between sequences
 - (c) Delete redundant data
 - (d) Encrypt sequences
9. Substitution matrices are used to:
 - (a) Store sequences
 - (b) Measure similarity between elements
 - (c) Visualize data
 - (d) Sort sequences
10. Probabilistic models are used to:

- (a) Eliminate uncertainty
 - (b) Represent data using probability
 - (c) Store large datasets
 - (d) Visualize graphs
11. BLAST is mainly used for:
- (a) Data compression
 - (b) Fast similarity search in biological databases
 - (c) Image processing
 - (d) Network analysis
12. Gene identification and annotation involves:
- (a) Deleting genes
 - (b) Identifying genes and understanding their function
 - (c) Storing sequences only
 - (d) Encrypting biological data
13. Protein structure analysis helps in:
- (a) Understanding protein function
 - (b) Reducing data size
 - (c) Sorting sequences
 - (d) Ignoring patterns
14. Biological data analysis is used in:
- (a) Disease diagnosis and drug discovery
 - (b) Only gaming
 - (c) Only networking
 - (d) Only graphics design
15. One advantage of sequence data mining is:
- (a) Ignoring patterns
 - (b) Revealing temporal and structural patterns
 - (c) Deleting data
 - (d) Limiting applications
16. Sequence data mining supports:
- (a) Random guessing

- (b) Prediction and decision making
 - (c) Data deletion
 - (d) Hardware optimization only
17. Sequence data mining is applied in:
- (a) Finance, healthcare, and bioinformatics
 - (b) Only education
 - (c) Only sports
 - (d) Only entertainment
18. Sequence data mining can handle:
- (a) Simple and small data only
 - (b) Complex and high-dimensional data
 - (c) Only text data
 - (d) Only images

True & False Questions

1. Biological sequence data includes DNA, RNA, and proteins.
2. DNA sequences are made of amino acids.
3. Protein sequences are made of amino acids.
4. The order of elements in biological sequences is not important.
5. Biological sequences are usually short and simple.
6. Biological data often contains hidden patterns.
7. Sequence alignment helps find similarities between sequences.
8. BLAST is a slow and inefficient tool.
9. Probabilistic models use probability to represent data.
10. Sequence data mining helps in prediction and decision making.

Answers

MCQs:

1. b
2. b
3. b
4. b
5. b
6. b
7. b
8. b
9. b
10. b
11. b
12. b
13. a
14. a
15. b
16. b
17. a
18. b

True & False:

1. True
2. False
3. True
4. False
5. False
6. True

7. True
8. False
9. True
10. True

7.7 Basic DateTime Operations in Python

7.7.1 Overview of the DateTime Module

Purpose of the Module

- Python has an in-built module named **DateTime** to deal with dates and times in numerous ways. In this article, we are going to see basic DateTime operations in Python.

Explanation:

time and date handle to module ready a provides Python
بسهولة بدون ما نكتب كل حاجة من
الصفء.

Basic Date and Time Handling in Python

Explanation:

Using this module, we can create, format, compare, and manipulate dates and times.

7.7.2 Main Classes in the datetime Module

There are six main object classes with their respective components in the datetime module mentioned below:

- `datetime.date`
 - `datetime.time`
 - `datetime.datetime`
 - `datetime.tzinfo`
 - `datetime.timedelta`
 - `datetime.timezone`
- Now we will see the program for each of the functions under datetime module mentioned above.

Explanation:

Each class is responsible for a specific part:

- date → التاريخ فقط
- time → الوقت فقط
- datetime → الاثنين مع بعض
- timedelta → الفرق بين وقتين

7.7.3 datetime.date()

We can generate date objects from the date class. A date object represents a date having a year, month and day.

Syntax

`datetime.date(year, month, day)`

`strftime` to print day, month and year in various formats. Here are some of them are:

- `current.strftime("%m/%d/%y")`
- `current.strftime("%b-%d-%Y")`
- `current.strftime("%d/%m/%Y")`
- `current.strftime("%B %d, %Y")`

Explanation:

`strftime` is used to format date in different shapes.

Example: month/day/year or full month name.

7.7.4 datetime.time()

A time object generated from the time class represents the local time.

Components:

- hour
- minute
- second
- microsecond
- tzinfo

Syntax

`datetime.time(hour, minute, second, microsecond)`

7.7.5 datetime.datetime()

datetime.datetime() module shows the combination of a date and a time.

Components:

- year
- month
- day
- hour
- minute
- second
- microsecond
- tzinfo

Syntax

datetime.datetime(year, month, day)

datetime.datetime(year, month, day, hour, minute, second, microsecond)

Current date and time using the strftime() method in different ways:

- strftime("%d")
- strftime("%m")
- strftime("%Y")
- strftime("%H:%M:%S")
- strftime("%m/%d/%Y, %H:%M:%S")

Explanation:

datetime is the most important class لأنه يجمع التاريخ والوقت مع بعض.

7.7.6 datetime.timedelta()

Definition and Meaning

It shows a duration that expresses the difference between two date, time or datetime instances to microsecond resolution.

Explanation:

Used to calculate الفرق بين تاريخين أو وقتين.

Past and Future Day Calculations

- Here we implemented some basic functions and printed past and future days.

Minimum and Maximum Values

- Also, we will print some other attributes of timedelta max, min that show maximum days and time, minimum date and time

Arithmetic Operations on Dates and Times

- Here we will also apply some arithmetic operations on two different dates and times.

7.7.7 Time Module Functions

time() method

The time() method returns the time as a floating-point number expressed in seconds since the epoch, in UTC

Syntax

```
time.time([])
```

ctime() method

ctime() method converts a time expressed in seconds since the epoch to a string representing local time.

Syntax

```
time.ctime([sec])
```

sleep() method

Python time method sleep() stops execution for the given number of seconds.

Syntax

```
time.sleep([sec])
```

localtime() method

localtime() method converts number of seconds to local time.

Syntax

```
time.localtime([sec])
```

gmtime() method

gmtime() method converts a time expressed in seconds since the Epoch to a struct_time in UTC.

Syntax

```
time.gmtime([sec])
```

mktime() method

It is the inverse function of localtime() method.

Syntax

```
time.mktime([t])
```

asctime() method

Python time method asctime() converts a struct_time representing a time to a 24-character string.

Syntax

```
time.asctime([t])
```

7.8 Assessment Questions

Multiple Choice Questions (MCQs)

1. The DateTime module in Python is used to:
 - (a) Handle images
 - (b) Work with dates and times
 - (c) Process text only
 - (d) Manage files
2. Using the datetime module, we can:
 - (a) Only store dates
 - (b) Create, format, compare, and manipulate dates and times
 - (c) Only delete dates
 - (d) Only print time
3. Which of the following is NOT a class in the datetime module?
 - (a) datetime.date
 - (b) datetime.time
 - (c) datetime.random

- (d) `datetime.timedelta`
- 4. The `datetime.date` class represents:
 - (a) Time only
 - (b) Date with year, month, and day
 - (c) Both date and time
 - (d) Time zone only
- 5. The correct syntax to create a date object is:
 - (a) `datetime.date(day, month, year)`
 - (b) `datetime.date(year, month, day)`
 - (c) `datetime.date(month, year, day)`
 - (d) `datetime.date(year, day)`
- 6. The `strptime()` function is used to:
 - (a) Compare dates
 - (b) Format date and time
 - (c) Delete time
 - (d) Add dates
- 7. A time object represents:
 - (a) Date only
 - (b) Local time
 - (c) Global time only
 - (d) Year only
- 8. Which of the following is NOT a component of time?
 - (a) Hour
 - (b) Minute
 - (c) Month
 - (d) Second
- 9. `datetime.datetime` represents:
 - (a) Only date
 - (b) Only time
 - (c) Combination of date and time
 - (d) Only timezone

10. Which class is used to calculate the difference between two dates?
- (a) `datetime.time`
 - (b) `datetime.timedelta`
 - (c) `datetime.date`
 - (d) `datetime.tzinfo`
11. `timedelta` is used to:
- (a) Store text
 - (b) Represent duration between dates or times
 - (c) Display images
 - (d) Sort data
12. The `time()` function returns:
- (a) Current date
 - (b) Time in seconds since epoch
 - (c) Formatted string
 - (d) Local time only
13. The `ctime()` function is used to:
- (a) Convert seconds to string time
 - (b) Pause execution
 - (c) Return UTC time
 - (d) Convert string to seconds
14. The `sleep()` function is used to:
- (a) Stop execution for a given number of seconds
 - (b) Convert time
 - (c) Format time
 - (d) Get system time
15. `localtime()` converts:
- (a) String to time
 - (b) Seconds to local time
 - (c) UTC to seconds
 - (d) Time to string
16. `gmtime()` returns:

- (a) Local time
 - (b) UTC time
 - (c) String format
 - (d) Random time
17. `mktime()` is:
- (a) Same as `gmtime()`
 - (b) Inverse of `localtime()`
 - (c) Used for formatting
 - (d) Used for delay
18. `asctime()` converts:
- (a) Seconds to integer
 - (b) `struct_time` to string
 - (c) String to `struct_time`
 - (d) Time to float

True & False Questions

1. The `datetime` module is built into Python.
2. `datetime.date` includes time information.
3. `strftime()` is used for formatting date and time.
4. `datetime.time` includes year and month.
5. `datetime.datetime` combines date and time.
6. `timedelta` represents the difference between two dates or times.
7. `time.time()` returns time in seconds since the epoch.
8. `sleep()` function speeds up execution.
9. `gmtime()` returns UTC time.
10. `mktime()` is the inverse of `localtime()`.

Answers

MCQs:

1. b
2. b
3. c
4. b
5. b
6. b
7. b
8. c
9. c
10. b
11. b
12. b
13. a
14. a
15. b
16. b
17. b
18. b

True & False:

1. True
2. False
3. True
4. False
5. True
6. True

7. True
8. False
9. True
10. True

7.9 Time Series Analysis and Visualization in Python

7.9.1 Introduction to Time Series Analysis

Definition of Time Series Data

- Time series data is information collected in sequence over time. It shows how things change at different points, like stock prices every day or temperature every hour.

Explanation:

Time series = data ordered by time.

Example: daily sales, hourly temperature.

Importance in Real-World Applications

- It is used in industries such as finance, pharmaceuticals, social media, and research.

Explanation:

Used in many fields because most real data depends on time.

Goal of Analysis and Visualization

- Analyzing and visualizing this data helps us find trends, seasonal patterns, and behaviors.
- These insights support forecasting and guide better decision-making.
- The main goal is to study data in time order to extract meaningful patterns and predictions.

Explanation:

We analyze time data to understand the past and predict the future.

7.9.2 Core Concepts in Time Series Analysis

Trend

- Trend: It represents the general direction in which a time series is moving over an extended period. It checks whether the values are increasing, decreasing or staying relatively constant.

Explanation:

Trend = overall direction (up / down / .(

Seasonality

- **Seasonality:** Seasonality refers to repetitive patterns or cycles that occur at regular intervals within a time series corresponding to specific time units like days, weeks, months or seasons.

Explanation:

Repeated patterns at fixed time (مثلاً كل سنة أو كل شهر).

Moving Average

- **Moving average:** It is used to smooth out short-term fluctuations and highlight longer-term trends or patterns in the data

Explanation:

Used to reduce noise and make trend clearer.

Noise

- **Noise:** It represents the irregular and unpredictable components in a time series that do not follow a pattern.

Explanation:

Random changes in data (no clear reason).

Differencing

- **Differencing:** It is used to make the difference in values of a specified interval. By default it's 1 but we can specify different values for plots.

Explanation:

We subtract consecutive values to remove trend and make data stable.

Stationarity

- **Stationarity:** A stationary time series is statistical properties such as mean, variance and autocorrelation remain constant over time.

Explanation:

Important concept: data should not change its behavior over time.

Order of Differencing

- **Order:** The order of differencing refers to the number of times the time series data needs to be differenced to achieve stationarity.

Explanation:

How many times we apply differencing (1, 2, ...).

Autocorrelation

- Autocorrelation: Autocorrelation is a statistical method used in time series analysis to quantify the degree of similarity between a time series and a lagged version of itself.

Explanation:

Compare current value with previous values.

Resampling

- Resampling: Resampling is a technique in time series analysis that is used for changing the frequency of the data observations.

Explanation:

Example: convert daily data $\rightarrow$ monthly data.

7.9.3 Types of Time Series Data

- Time series data can be classified into two sections:

Continuous Time Series

- Continuous Time Series: Data recorded at regular intervals with a continuous range of values like temperature, stock prices, Sensor Data, etc.

Explanation:

Values are numeric and continuous.

Discrete Time Series

- Discrete Time Series: Data with distinct values or categories recorded at specific time points like counts of events, categorical statuses, etc.

Explanation:

Values are counts or categories.

7.9.4 Visualization Approaches**Line Plots and Area Charts**

- Use line plots or area charts for continuous data to highlight trends and fluctuations.

Explanation:

Best for showing changes over time.

Bar Charts and Histograms

- Use bar charts or histograms for discrete data to show frequency or distribution across categories.

Explanation:

Used for counts and comparisons.

7.10 Assessment Questions**Multiple Choice Questions (MCQs)**

1. Time series data is defined as:
 - (a) Random data
 - (b) Data collected over time in sequence
 - (c) Only categorical data
 - (d) Static data
2. An example of time series data is:
 - (a) Student names
 - (b) Daily stock prices
 - (c) Colors list
 - (d) File sizes
3. Time series analysis is widely used in:
 - (a) Finance and research
 - (b) Only gaming
 - (c) Only networking
 - (d) Only graphics
4. The goal of time series analysis is to:
 - (a) Delete data
 - (b) Find patterns and make predictions
 - (c) Store data only
 - (d) Encrypt data
5. Trend represents:

- (a) Random noise
 - (b) General direction of data over time
 - (c) Data storage
 - (d) Data format
6. Seasonality refers to:
- (a) Random variation
 - (b) Repeating patterns at fixed intervals
 - (c) Increasing trend only
 - (d) Decreasing trend only
7. Moving average is used to:
- (a) Increase noise
 - (b) Smooth short-term fluctuations
 - (c) Delete data
 - (d) Store values
8. Noise in time series represents:
- (a) Regular patterns
 - (b) Predictable behavior
 - (c) Irregular and random variations
 - (d) Trends
9. Differencing is used to:
- (a) Add values
 - (b) Subtract consecutive values
 - (c) Multiply values
 - (d) Sort data
10. Stationarity means:
- (a) Data changes randomly
 - (b) Statistical properties remain constant over time
 - (c) Data is deleted
 - (d) Data is sorted
11. Order of differencing refers to:
- (a) Number of variables

- (b) Number of times differencing is applied
 - (c) Size of dataset
 - (d) Type of data
12. Autocorrelation measures:
- (a) Difference between datasets
 - (b) Similarity between a series and its lagged values
 - (c) Data size
 - (d) Frequency of data
13. Resampling is used to:
- (a) Delete data
 - (b) Change frequency of observations
 - (c) Sort data
 - (d) Compress files
14. Continuous time series data is:
- (a) Categorical only
 - (b) Numeric and continuous values
 - (c) Random data
 - (d) Text data
15. Discrete time series data is:
- (a) Continuous values
 - (b) Counts or categories
 - (c) Always numeric
 - (d) Always random
16. Line plots are best used for:
- (a) Discrete data only
 - (b) Continuous data over time
 - (c) Text analysis
 - (d) File comparison
17. Bar charts are useful for:
- (a) Continuous trends
 - (b) Frequency of categories

- (c) Time smoothing
- (d) Noise reduction

True & False Questions

1. Time series data is ordered by time.
2. Time series analysis is not useful in real-world applications.
3. Trend shows the overall direction of data.
4. Seasonality occurs at irregular intervals.
5. Moving average helps smooth data.
6. Noise represents predictable patterns.
7. Differencing helps make data more stable.
8. Stationarity means statistical properties change over time.
9. Autocorrelation compares current and past values.
10. Resampling changes the frequency of data.

Answers

MCQs:

1. b
2. b
3. a
4. b
5. b
6. b
7. b
8. c
9. b

10. b
11. b
12. b
13. b
14. b
15. b
16. b
17. b

True & False:

1. True
2. False
3. True
4. False
5. True
6. False
7. True
8. False
9. True
10. True